Weekly Literature Review

Period: Nov 06 – Nov 13, 2026

This Week in Brief

The week's biggest story is two high-profile atherosclerosis failures: pelacarsen did not reduce events despite large, sustained Lp(a) lowering, and ziltivekimab gave no MACE benefit despite deep IL-6 and hsCRP suppression, a setback for biomarker-led drug development. The wins came from established tools: left bundle-branch pacing cut the composite of death, HF hospitalization, or pacing-induced cardiomyopathy versus RV pacing in AV block, and a systolic target below 120 mm Hg lowered events after ischemic stroke. For transplant practice, dapagliflozin modestly protected kidney function in heart transplant recipients without an infection signal.

Cardiology

Two flagship next-target atherosclerosis trials failed this week: lipoprotein(a) lowering with pelacarsen and IL-6 blockade with ziltivekimab, despite large biomarker reductions. That is a sobering result for biomarker-led drug development. The positive trials used established tools: a lower blood pressure target after stroke and conduction system pacing in place of RV pacing. Extended ISCHEMIA follow-up again showed no survival gain from routine invasive management.

Pelacarsen and Cardiovascular Outcomes in Patients with High Lipoprotein(a)

N Engl J Med · Cho L, Nicholls SJ, Frndak S, et al. · Placebo-controlled RCT, 8,350 patients · DOI

Area: Lipoprotein(a) lowering, secondary prevention

Design: Patients with established cardiovascular disease and Lp(a) of at least 70 mg/dL were randomized to monthly subcutaneous pelacarsen 80 mg or placebo. Primary endpoint: CV death, nonfatal MI, nonfatal stroke, or urgent coronary revascularization. Median follow-up was 4.7 years.

Results: Median baseline Lp(a) was 236 nmol/L; at 12 months it was 64.4 nmol/L with pelacarsen vs 233.4 nmol/L with placebo. The primary endpoint occurred in 12.3% vs 12.1% (HR 1.02, 95% CI 0.90 to 1.15; P=0.76). In patients with Lp(a) of at least 90 mg/dL, rates were 12.3% vs 12.3% (HR 1.00; 95% CI 0.87 to 1.15). Serious adverse events occurred in 37.3% vs 41.1%.

Context: The first Lp(a) outcomes trial is flatly null, even in the highest-Lp(a) stratum, despite large and sustained Lp(a) reductions over nearly 5 years. Genetic data had made Lp(a) look causal, though Mendelian randomization implied that very large absolute reductions would be needed for modest benefit. This result raises the bar sharply for the siRNA programs (olpasiran, lepodisiran) and oral muvalaplin. For now, Lp(a) remains a risk enhancer to measure, not a treatment target.

Interleukin-6 Inhibition with Ziltivekimab for Atherosclerotic Disease

N Engl J Med · Ridker PM, Lincoff AM, Marx N, et al. · Double-blind RCT, 6,376 patients · DOI

Area: IL-6 inhibition in ASCVD with CKD

Design: Patients with ASCVD, CKD, and hsCRP of at least 2 mg/L were randomized to monthly subcutaneous ziltivekimab 15 mg or placebo. Primary endpoint: 3-point MACE (CV death, nonfatal MI, nonfatal stroke). Median follow-up was 33 months.

Results: At 24 months, the placebo-adjusted median change was -76 percentage points for free IL-6 and -64 percentage points for hsCRP. MACE occurred in 17.6% vs 17.7% (HR 0.99; 95% CI 0.88 to 1.11; P=0.84). Serious adverse events were 54.6% vs 54.8%; serious infections were 20.4% vs 17.9%.

Context: Profound IL-6 and CRP suppression produced no MACE benefit in the population thought most likely to respond: high residual inflammatory risk with CKD. That undercuts the reasoning extended from CANTOS that hitting the IL-1β/IL-6 axis would generalize. It also shows that hsCRP lowering is not a valid surrogate for benefit. The infection excess argues against off-label enthusiasm. Ziltivekimab's HFpEF program (HERMES) now carries more weight for the IL-6 hypothesis.

Left Bundle-Branch Pacing versus Right Ventricular Pacing for Atrioventricular Block

N Engl J Med · Qian Z, Jiang L, Hou X, et al. · Open-label RCT, 702 patients · DOI

Area: Conduction system pacing for AV block

Design: Patients with AV block were randomized 1:1 to left bundle-branch pacing or RV pacing at international centers. Primary endpoint: death, HF hospitalization, or pacing-induced cardiomyopathy. The trial was stopped early at a prespecified interim analysis; median follow-up was 35.1 months.

Results: In the modified ITT population of 685, primary events occurred in 16 vs 35 patients. Four-year cumulative incidence was 8.5% vs 19.2% (HR 0.41; 95% CI 0.23 to 0.75; P=0.003). Serious adverse events unrelated to implantation were 11.8% vs 9.3% (P=0.32).

Context: This is the first randomized hard-outcome evidence that left bundle-branch pacing beats RV pacing in AV block, with the composite risk cut by more than half. Prior support for conduction system pacing was largely observational or echo-based. Three caveats limit it: few events, early stopping that likely inflates the effect, and an open-label design with imaging-defined pacing cardiomyopathy that is vulnerable to ascertainment bias. Component results were not reported in the abstract. Still, this should push LBBP toward default status when high pacing burden is expected.

Intensive Blood-Pressure Control in Patients with Previous Ischemic Stroke

N Engl J Med · Sampaio Silva G, Machline-Carrion MJ, Santo K, et al. · Open-label RCT, 4,368 patients · DOI

Area: Blood pressure target after ischemic stroke/TIA

Design: Patients at 30 Brazilian sites with prior ischemic stroke or TIA and elevated BP were randomized to a systolic target below 120 or below 140 mm Hg. Primary endpoint: CV death, nonfatal MI, nonfatal

stroke, or hospitalization for unstable angina or HF. Median follow-up was 3.9 years.

Results: Achieved SBP was 122.0 vs 135.2 mm Hg (difference 13.1). The primary endpoint occurred in 10.9% vs 13.5% (3.25 vs 4.11 per 100 person-years; HR 0.79; 95% CI 0.67 to 0.94; P=0.007). The composite renal outcome was more common with intensive treatment; overall serious adverse events were similar.

Context: Targeting SBP below 120 after ischemic stroke or TIA cut major CV events by about a fifth, closing a gap left by SPRINT, which excluded stroke, and by underpowered prior stroke BP trials. Its sister trial, OPTIMAL-DIABETES, published alongside it, was null in type 2 diabetes. That echoes ACCORD-BP and conflicts with China's positive BPROAD, so the diabetes question stays open. The renal signal mirrors SPRINT and calls for eGFR monitoring. This supports a lower target after cerebrovascular events.

Dapagliflozin for Renal Protection in Heart Transplant Recipients: A Randomized Clinical Trial

Circulation · Holmen S, Manintveld O, Damman K, et al. · Double-blind RCT, 430 patients · DOI

Area: SGLT2 inhibition after heart transplantation

Design: Adult heart transplant recipients at least 1 year post-transplant with eGFR of at least 25, at six Northern European centers, were randomized to dapagliflozin 10 mg or placebo. Primary endpoint: eGFR slope from 2 weeks to the end of 12 months of blinded treatment.

Results: Baseline eGFR was 67 mL/min/1.73 m², and median time from transplant was 7 years. eGFR changed by 2.0 with dapagliflozin vs 0.3 with placebo; the difference was 1.7 (95% CI 0.1 to 3.3; p=0.03). Adverse event rates, including infections, were similar, with no unexpected safety signals.

Context: This is the first adequately sized randomized evidence that dapagliflozin modestly protects kidney function in heart transplant recipients. Calcineurin-inhibitor nephropathy drives much late morbidity in this population, which the major SGLT2 kidney trials excluded. Effect size is small, the endpoint is a 12-month surrogate, and the confidence interval approaches zero. The infection safety data in immunosuppressed patients are probably the most practice-relevant finding. Routine use in stable recipients, especially those with CKD or diabetes, looks reasonable while hard renal outcomes remain unproven.

Also notable

- Cangrelor versus crushed ticagrelor in acute myocardial infarction complicated by cardiogenic shock (DAPT-SHOCK-AMI): a randomised, double-blind trial (*Lancet*). In AMI cardiogenic shock, cangrelor achieved platelet inhibition far more often than crushed ticagrelor, but 30-day noninferiority failed (38% vs 41% death/MI/stroke).
- Hemodynamic effects of semaglutide in individuals with heart failure with preserved ejection fraction and obesity: a placebo-controlled randomized clinical trial (*Nat Med*). In 81 obese HFpEF patients, semaglutide 2.4 mg lowered exercise wedge pressure 10.6 mm Hg versus placebo; hemodynamic remission occurred in 53.7% versus 7.4%.
- Discontinuing Statins in Older Adults without Cardiovascular Disease (*N Engl J Med*). Stopping primary-prevention statins at age 70 or older with multimorbidity met noninferiority, though events were 12.7% versus 9.2% with continuation.

AI in Medicine

Cardiovascular AI is moving into randomized and large external validation, and the gains are smaller than the retrospective AUCs suggested. The randomized AI-ECG atrial fibrillation (AF) screening trial found that flagging high-risk ECGs raised diagnostic yield but gave a statistically fragile population effect. AI-derived coronary inflammation from routine calcium-score CT and large language models (LLMs) used to audit care gaps point to opportunistic, data-reuse applications as the nearer-term wins.

Leveraging AI-ECG Technology for Early Notification and Tracking of AF Development (LATENT): A Randomized Controlled Trial

Circulation · Lee CC, Lin WY, Liu WT, et al. · Open-label single-center RCT, 14,665 patients · DOI

Area: AI-ECG for atrial fibrillation screening

Design: Adults 40 to 85 without prior AF or flutter were randomized to an AI-ECG-guided strategy or usual care. High-risk sinus-rhythm ECGs prompted an offer of Holter monitoring for up to 7 days, and matched controls received identical monitoring. Primary endpoint: new AF, flutter or pre-AF atrial findings within 90 days.

Results: Only 84 (1.1%) intervention and 78 (1.1%) control participants were high risk; 48 (57.1%) of the high-risk intervention group were monitored. The primary outcome occurred in 54 (0.74%) vs 35 (0.48%) participants (OR 1.54, 95% CI 1.01-2.37; P=0.046); reclassifying 1 event moved P to 0.057. Yield was 32.1% vs 2.6% among high-risk participants. AF/flutter detection (0.3% vs 0.2%) and antiarrhythmic initiation (2.3% vs 2.4%) did not differ. Monitoring adverse events: 22.9% vs 18.8%.

Context: An AI-ECG flag strongly enriches for atrial pathology in the individuals it identifies, but as a population strategy the benefit was marginal and rested mostly on frequent ectopy and nonsustained runs rather than clinical AF. That fits prior nonrandomized AI-guided screening and the broader screening literature, where finding more arrhythmia has not reliably translated into stroke prevention. Low flag rates, 57% monitoring uptake and a single-center design cap the effect. Treat this as a targeting tool awaiting outcome trials, not a screening program to adopt.

Assessment of Coronary Inflammatory Risk From Noncontrast Cardiac Computed Tomography

JAMA Cardiol · Chan K, Gissler MC, Sobirov I, et al. · Multicenter cohort with external validation, 25,928 patients · DOI

Area: AI coronary inflammation (FAI) on noncontrast CT

Design: An AI algorithm measuring the coronary fat attenuation index (FAI) on noncontrast calcium-score CT was trained in 6818 patients and tested in an external cohort of 19,110 who had clinically indicated CT. It was compared with CCTA-derived FAI and the Agatston score. Primary outcome: CV mortality; secondary: MACE. Median follow-up 6.7 years.

Results: There were 536 CV deaths and 1727 MACE. Adding CCTA FAI or noncontrast FAI to the CAC score raised the AUC from 0.76 to 0.81 and 0.80 (P < .001), with improved reclassification and net benefit. With CAC of 0, noncontrast FAI above the 75th percentile raised CV mortality from 1.4 to 4.0 per 1000 person-years and MACE from 5.9 to 13.3. Elevated noncontrast FAI was present in 44.9% of CV deaths with CAC 0 and 44.2% with CAC 1 through 99.

Context: Coronary inflammation can now be read from the cheap, ubiquitous calcium-score scan, with discrimination nearly matching CCTA-based FAI. That matters most where a zero calcium score currently ends the conversation, since nearly half of CV deaths in that group had high inflammatory signal. It extends the Oxford perivascular fat work from CCTA to noncontrast CT. However, it is observational and proprietary, and no trial shows that intensifying statins or adding colchicine on the basis of FAI improves outcomes. For now, treat it as a risk enhancer and a candidate trial-enrichment tool.

AI-Identified Reasons for Nonescalation of Lipid-Lowering Therapy for Secondary Prevention

JAMA Cardiol · Retrospective cohort, 28,113 patients · LLM extraction of lipid-therapy inertia

An LLM read clinic notes for ASCVD patients with LDL-C above goal at a Northern California system, with F1 0.80 to 0.98 against annotators. Therapy was not escalated in 88.7% of notes, most often citing patient-reported adverse effects (12.6%). This is single-system and documentation-dependent, and no intervention was tested.

ClinicalAge, a framework for modelling ageing trajectories from longitudinal electronic health records in the UK: a model development and validation study

Lancet Digit Health · Model development and external validation, about 5 million patients · EHR-derived biological aging clock

A transformer trained on UK Biobank EHR data to approximate PhenoAge (R2 0·819) was validated in 4 728 376 CPRD patients. It was associated with all-cause mortality (adjusted HR 1·23 UK Biobank; 1·36 CPRD) and midlife divergence with diabetes and hypertension. It is anchored to a proxy clock and remains associational.

AI in Medical Education

This week's most useful AI-education work measured performance rather than attitudes. A multicenter randomized trial found that AI feedback improves clinician accuracy but that novices are more easily misled by wrong advice. A national survey showed US internal medicine clerkships still largely lack AI policies and secure tools. The rest of the week was mostly single-site perception surveys, frameworks, and chatbot exam benchmarking.

AI feedback for clinician assessment of umbilical and middle cerebral artery Doppler ultrasound: a multicenter randomized trial

Am J Obstet Gynecol MFM · Bashir Z, Wong CK, Lin M, et al. · Multicenter RCT, 96 clinicians · DOI

Area: AI feedback for obstetric Doppler assessment

Design: At five Danish hospitals, 96 OB/GYN clinicians (residents through obstetricians), paired by ultrasound experience, were randomized 1:1 to AI visual and textual feedback or none. Each judged 40 static umbilical and middle cerebral artery Doppler vignettes for gate placement and waveform adequacy. Accuracy was scored against expert consensus.

Results: Accuracy was 75.0% with AI feedback vs 62.3% without (mean difference 12.7%; 95% CI, 8.2-17.1; $p < 0.001$). In novices (n=36) it was 70.0% vs 57.9% (12.1%; 95% CI 2.4-21.7). In experienced clinicians (n=60) it was 78.0% vs 65.0% (13.0%; 95% CI 5.5-20.5). Novices followed incorrect AI feedback more often than experienced clinicians (22.3% vs 15.8%; P = .01). AI alone outperformed unassisted novices, while experienced clinicians with AI outperformed AI alone.

Context: AI feedback raised accuracy by about 13 points at every experience level, but novices were likelier to adopt wrong suggestions. That is the automation-bias gradient that reader studies in imaging have repeatedly described. The fact that experienced clinicians with AI beat AI alone is the strongest argument here for human-AI teaming over replacement. This measures assisted performance on static images, not real-time scanning and not retained unassisted skill. It therefore cannot say whether AI feedback teaches trainees or simply substitutes for their judgment, which is the question educators need answered.

Policy, ethical, and practical considerations of artificial intelligence use in the internal medicine clerkship: results of a survey of clerkship directors

Acad Med · National survey, 114 US medical schools · AI policy in internal medicine clerkships

In this survey of IM clerkship directors (80.9% response), 39.5% reported no formal AI policy and 34.2% were unsure whether one existed. 73.7% would encourage student AI use for some clinical skill, but only 6.1% had firewall-protected tools. Data are descriptive and from late 2024.

Longevity Medicine

A thin week for human geroscience. The most useful human data point is a large transplant-registry analysis finding no longevity signal from chronic rapamycin. A new oral senolytic, a mineral supplement, extended lifespan only in male mice. Serial-sample sequencing showed how somatic clones expand in blood for decades before leukemia. Biological-age work stayed observational and correlative: epigenetic clocks, gait age and frailty scores.

Rapamycin and human longevity among kidney transplant recipients surviving the first year

Geroscience · Registry cohort, 11,474 kidney transplant recipients · Rapamycin as geroprotector

Among 1-year kidney transplant survivors followed a median 11 years, propensity-weighted median survival age was 64.6 years with rapamycin and 64.3 years without (difference 95% CI -1.6, 2.0 years). Caveat: immunosuppressive daily dosing in a transplant population, not low-dose intermittent geroprotective regimens.

Altered protein cysteine reactivity in senescent cells is a vulnerability for senolysis to ameliorate aging

Nat Aging · Preclinical mouse and cell study · Senolytics

Oral vanadyl sulfate exploited low glutathione reductase and glutathione disulfide buildup in senescent cells to trigger protein misfolding and selective death. It improved several age-related phenotypes and extended lifespan in male mice. Caveat: male mice only, no effect sizes reported, and vanadium toxicity is untested.

Evolutionary dynamics in the years preceding acute myeloid leukaemia

Nature · Serial-sample duplex sequencing cohort, pre-AML · Clonal hematopoiesis and somatic evolution

In annual blood samples from as early as 15 years before AML diagnosis, first drivers arose in early decades of life. Later multi-driver clones carried selective advantages up to 100-fold larger, and complete clonal sweeps preceded AML by decades. Caveat: descriptive, with no clinical risk tool yet.

General Medicine

Three themes stand out. Incretin-era obesity therapy keeps moving toward specific complications and body composition: retatrutide eased knee osteoarthritis pain alongside roughly 20% weight loss, oral orforglipron lowered apnea-hypopnea index in sleep apnea, and two trials addressed muscle loss, with myostatin/activin A blockade (COURAGE) and supervised resistance training both blunting lean-mass loss on GLP-1-based drugs. Perioperative practice got two large trials. PEPPER found low-dose aspirin was not noninferior to rivaroxaban or warfarin for VTE prophylaxis after hip or knee replacement, and a double-blind trial showed that holding the morning ACE inhibitor or ARB cut intraoperative hypotension in older adults without added harm. In oncology, personalized mRNA neoantigen therapy plus pembrolizumab became the first individualized cancer vaccine to improve recurrence-free survival in a phase 3 trial, in resected melanoma. A large claims cohort also linked first-trimester doxycycline to higher pregnancy loss.

Prevention of Venous Thromboembolism after Total Hip or Knee Replacement

N Engl J Med · Pragmatic noninferiority RCT, 18,883 patients · VTE prophylaxis after arthroplasty

PEPPER: aspirin 81 mg twice daily failed noninferiority to rivaroxaban and warfarin for death or symptomatic VTE (1.26% vs 0.94% vs 1.23%), with numerically fewer safety events.

Effect of discontinuing antihypertensives on perioperative hypotension in older patients before non-cardiac surgery: multicentre, randomised, double blind, controlled trial

BMJ · Double-blind placebo-controlled RCT, 2036 patients · Perioperative ACE inhibitor/ARB management

In older adults having noncardiac surgery, replacing the morning ACE inhibitor or ARB dose with placebo cut perioperative hypotension from 45.5% to 31.2%, with no change in 30-day complications.

Intismeran Autogene plus Pembrolizumab for Resected Stage IIB to IV Melanoma

N Engl J Med · Phase 3 double-blind RCT, 1137 patients · Adjuvant melanoma, personalized mRNA neoantigen therapy

In resected stage IIB to IV melanoma, adding intismeran to pembrolizumab raised 24-month recurrence-free survival to 82.2% from 70.0%; the benefit emerged only after 6 months.

Retatrutide Once Weekly in Adults with Knee Osteoarthritis Pain and Obesity

N Engl J Med · Two phase 3 RCTs, 445 and 574 patients · Knee osteoarthritis with obesity

In TRIUMPH-4 and TRIUMPH-1-OA, retatrutide reduced WOMAC knee pain more than placebo at every dose, with weight loss up to -23.7% at 68 weeks; serious adverse events were more common.

Trevogrumab (anti-GDF8) with or without garetosmab (anti-activin A) for prevention of muscle loss during semaglutide-induced weight reduction in people with obesity (COURAGE): a two-part, phase 2, double-blind, placebo-controlled, randomised trial

Lancet · Phase 2 double-blind RCT, 975 patients · Muscle preservation during GLP-1 weight loss

In COURAGE, adding trevogrumab plus garetosmab to semaglutide reduced lean-mass loss (+4.5% vs placebo) and deepened fat loss. Phase 2 data with no functional outcomes, and the abstract truncates safety.

Verification Notes

- 501 candidate papers screened from PubMed; 19 selected (8 with a full write-up).
- Every selected PMID, title, author list and journal is taken directly from PubMed. Design and Results lines are drawn from the abstract and their numbers checked against it automatically. Context lines reflect the model's knowledge of the prior literature and are not machine-verified.
- AI in Medical Education, PMID 42994530: the write-up cites 96, not found verbatim in the PubMed abstract. Check against the paper.

Selected and written by claude-opus-5-5 from PubMed records added Nov 06 – Nov 13, 2026.